

Climate Time-Series Sorting and Ranking

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START WITH THE SHAPE OF THE DATA

Start with a time series, not a leaderboard

A time series keeps each measurement attached to its date. Read climate measurements in time order first: date on one axis, a variable such as temperature anomaly on the other.





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THE QUESTION MOVES WITH THE ORDER

Sorting changes the question

The same values can support different questions.

Chronological order

When did values change?

Ascending order

Which periods were coolest?

Descending order

Which periods were warmest?

Keep the original dates attached to every value.

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A RANK IS A POSITION

Ranking turns values into positions

Sort the values, assign positions, and map each position back to its year or month. A rank says where a value sits in the selected set.

Year	Anomaly	Rank
2018	+0.72°C	4
2019	+0.95°C	2
2020	+1.02°C	1
2021	+0.84°C	3

Example climate ranking · highest anomaly receives rank 1.



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DEFINE THE SET FIRST

Choose the comparison window before ranking

Single year

Rank one year against other years.

Same month

Rank months against the same month across years.

Rolling average

Rank averaged values to reduce short-term noise.

NOAAGlobalTemp provides monthly and annual time series, including anomalies relative to a 1991–2020 climatology.

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RANKING IS SELECTIVE

Ranks reveal extremes, but not the whole trend

A top-ranked year may be an extreme point. Pair the ranking with the line chart, baseline, uncertainty, and the length of the comparison window.

Rank locates

The unusual period in the selected set.

The line explains

The broader pattern, context, and limits.



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MAKE THE TRANSFORMATION REPEATABLE

Use a reproducible sorting workflow

01

Clean

Dates, units, and missing values.

02

Define

Variable and comparison window.

03

Sort

Values while preserving timestamps.

04

Check

Ranks and ties.

05

Show

Series and ranking together.

A clear method makes the result easier to repeat, inspect, and explain.

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THE TAKEAWAY

Ranking is a lens, not a conclusion



Use ranks to locate unusual periods.



Use time order to explain change.



Use both to support a climate question, not replace one.

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